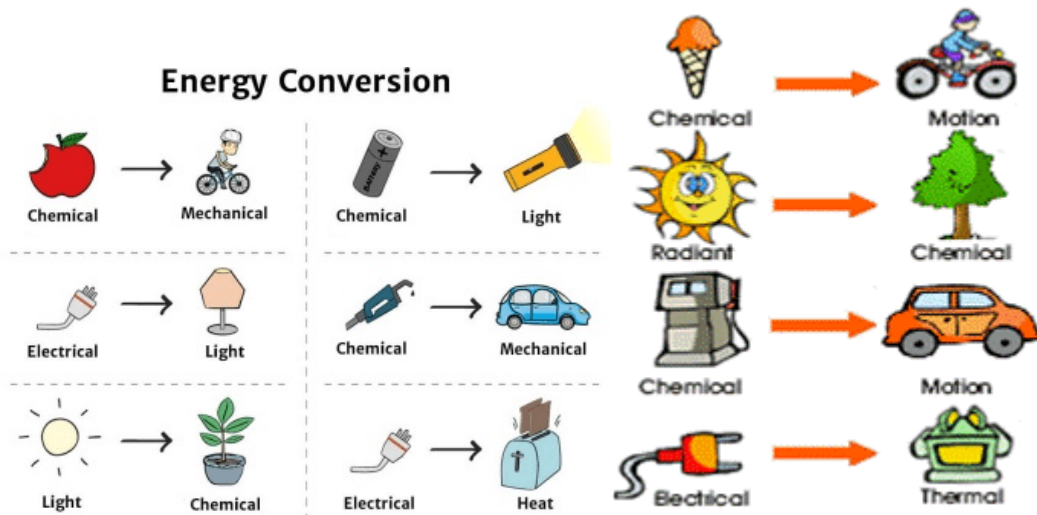


KWA-THEMA SKILL SCHOOL

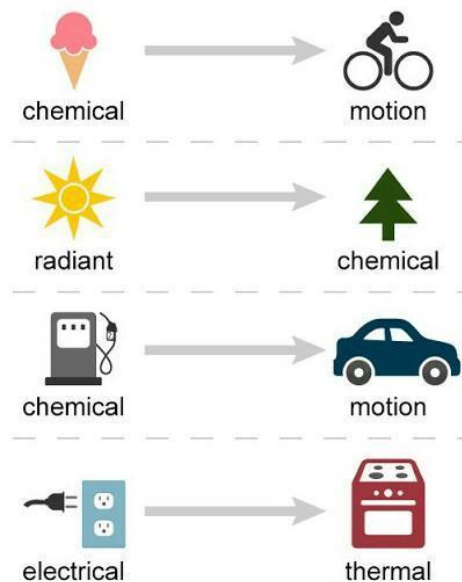
NS TECH

YEAR 4 TERM 3

ENERGY AND CHANGE



Energy transformations



NAME: _____

YEAR: _____

ELECTRIC CIRCUITS

1.1 STATIC ELECTRICITY:

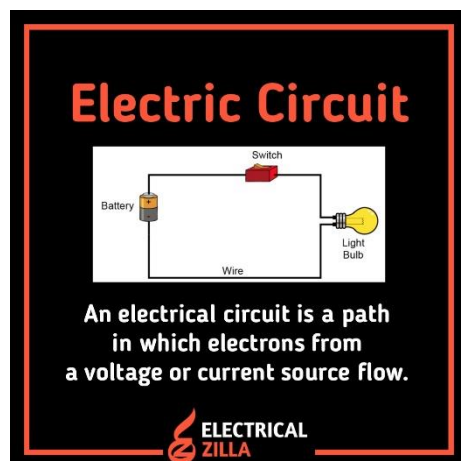
Imbalance between + and- charges in a material



What is Static Electricity?

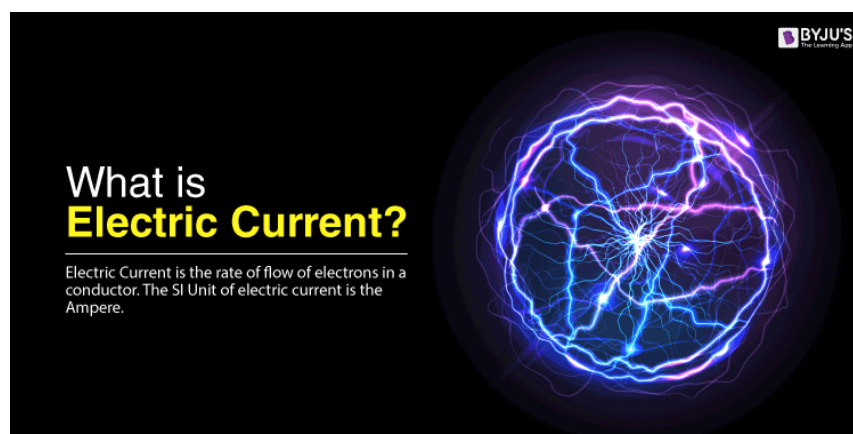
- Static electricity is the build up of an electric charge on an object.
- The word static means “not moving” so static electricity, unlike other electricity does not move through wires.
- If a large enough charge builds up however, it can “jump” from one object to another.
- When static electricity “jumps” it is called an electric discharge.

1.2 ELECTRIC CIRCUIT

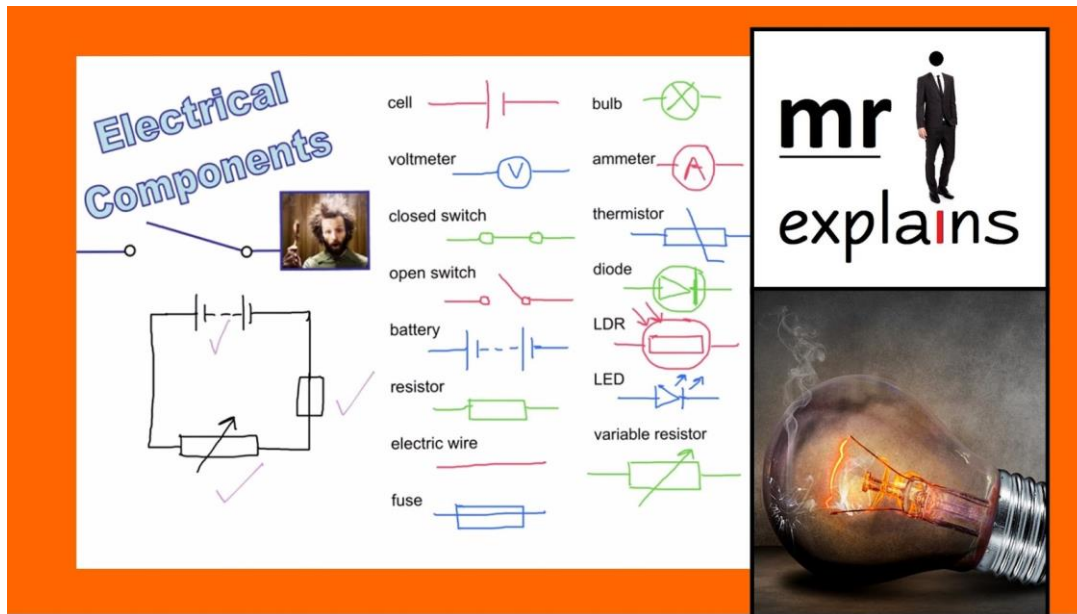


Electric circuit: The path for transmitting electric current

1.3 CURRENT ELECTRICITY

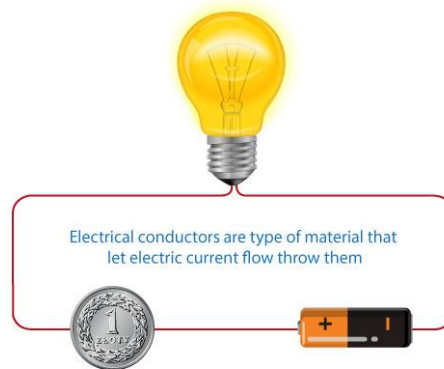


1.4 ELECTRIC COMPONENTS



1.5 CONDUCTORS

Electrical Conductor

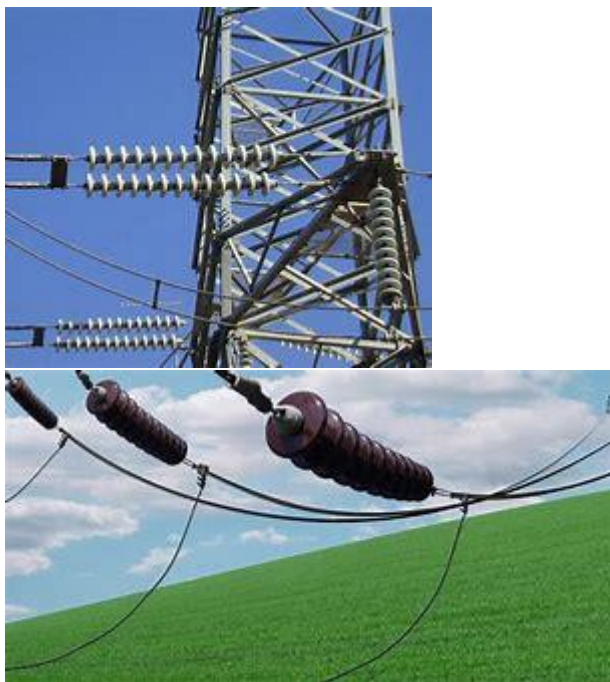
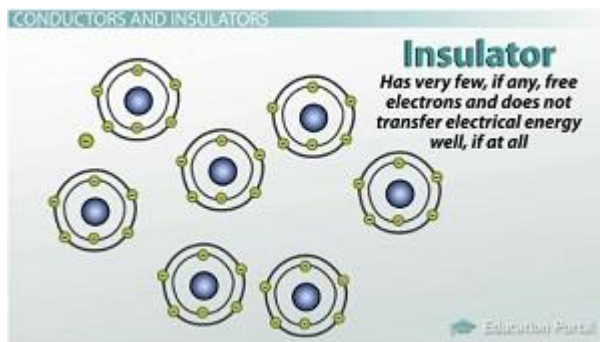


MPS
My Physics

Conductors: those substances that allow electricity to flow



1.6 INSULATOR

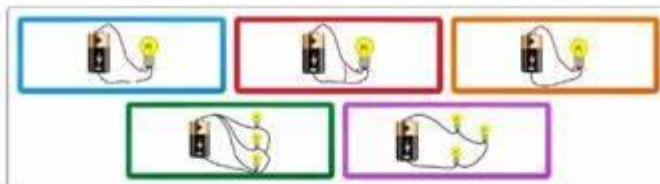


Insulators: those substance which do not allow electricity to flow

Conductors and insulators.

Conductors	Insulators
1. Those substances through which electricity can flow are called conductors.	1. Those substances through which electricity cannot flow are called insulators.
2. Electrical resistances of conductors are very low.	2. Electrical resistances of insulators are infinitely very high.
3. They contain large number of free electrons.	3. They do not contain free electrons.
4. Generally metals are conductors. E.g. silver, copper, aluminium	4. Generally non – metals are insulators. E.g. wood, rubber, plastic

1.1 ACTIVITIES: Look at the different pics and colors (blue A, B red, C orange green purple then indicate which one is short, parallel, open, close, series (5)



- ☐ Short circuit
- ☐ Parallel circuit
- ☐ Open circuit
- ☐ Close circuit
- ☐ Series circuit

MATCH COLUMN A WITH B MATCH CONCEPTS WITH CORRECT DESCRIPTION YOU MAY CHOOSE FROM THE BOX

Those substance which do not allow electricity to flow.
 Those substances which allows electricity to flow
 The path for transmitting the electric current
 Imbalance between positive and negative charges in a material
 Those are the electrical units cell, bulb, wire, fuse etc.

COLUMN A CONCEPTS	COLUMN B MATCHING DESCRIPTION
1.Conductors	
2.Electric components	
3.Static electricity	
4.Electric circuit	
5.Insulators	

(10)

ACTIVITY B

Matching type: Match Column A with B write the letter of the correct answer on the space provided before each number

A

B

_____ **1 Dry cell**

a. Path taken by electric current

_____ **2. Conductor**

**b. Path were electric flows from
Source back**

_____ **3. Light bulb**

**c. The copper wire where current
Flows**

_____ **4. Complete circuit**

**d.Contains material that allows
Chemical reaction**

_____ **5.Electric circuit**

e.Device that uses current

(10)

ACTIVITY: ON EXPANED OPPORTUNITIES: LEARNERS TO COMPARE DIFFERENCES BETWEEN INSULATORS AND CONDUCTORS (10)

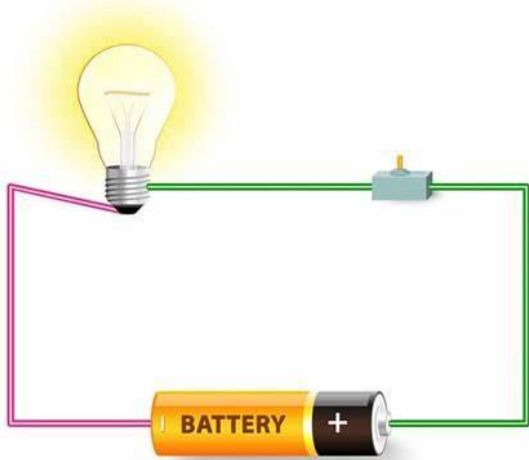
DIFFERENCES		MARK ALLOC
INSULATORS	CONDUCTORS	
1.	1.	
2.	2.	
3.	3.	
4.	4.	
5.	5.	

ACTIVITY: A SIMPLE CIRCUIT

INSTRUCTIONS

Look at the example of simple circuit. And answer the questions that follow:

SIMPLE ELECTRIC CIRCUIT



1.What are the components that made up this system for transferring electrical energy ?

2.Do you think this is an open or close circuit?

--

3.Which part is providing source of energy?

--

4.What is conducting material?

--

5.What is the output of this system?

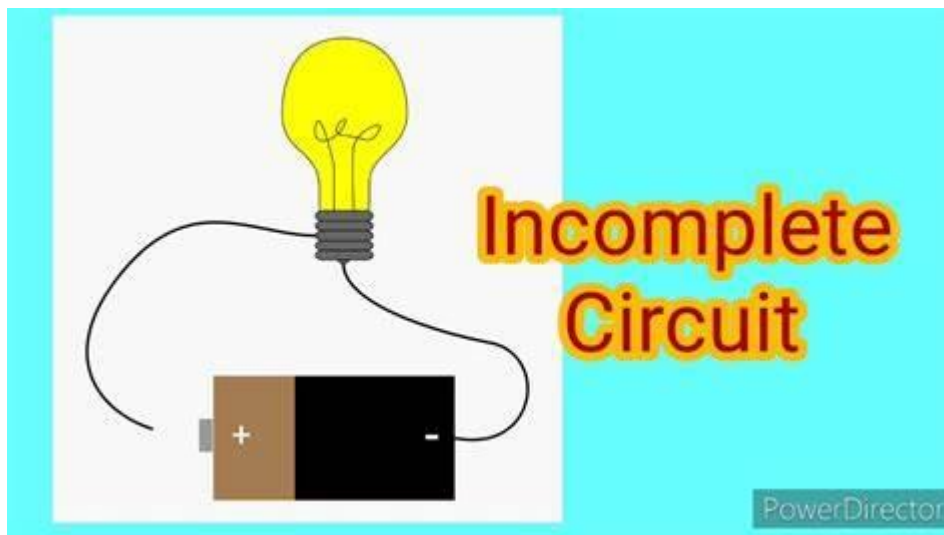
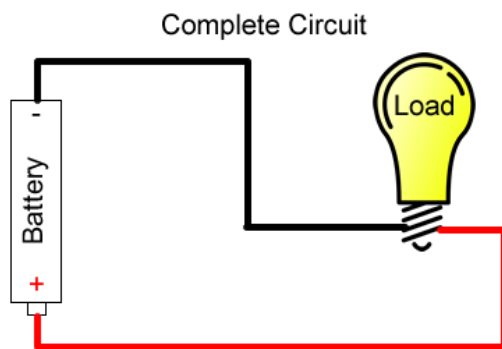
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(7)

EXPLAIN WHEN CIRCUIT IS COMPLETE

An electrical circuit is complete only when there is **at least one closed loop from the positive to the negative end**. This is the simplest form of an electric circuit. The circuit found inside a television is more complicated and has different components.

An electric **circuit** involves the flow of charge in a complete conducting loop. When here is an electric circuit light bulbs light, motors run, and a compass needle placed near a wire in the circuit will undergo a deflection. When there is an electri

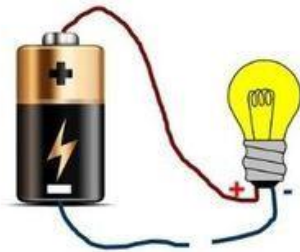


B.SYSTEM TO SOLVE PROBLEMS

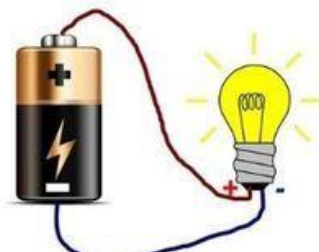
B.1 IDENTIFYING DIFFERENT CIRCUITS

**BRIEFLY DEFINE FF CONCEPTS SERIES,PARALLEL
COMPARE THE STRENGTH OF CURRENT OF CELLS**

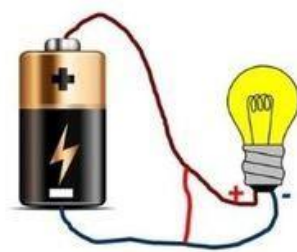
Types of Electric Circuits



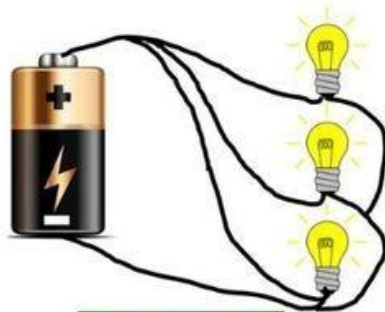
Open Circuit



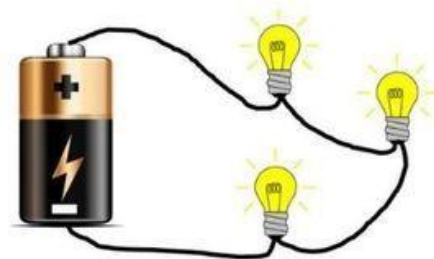
Close Circuit



Short Circuit



Parallel Circuit

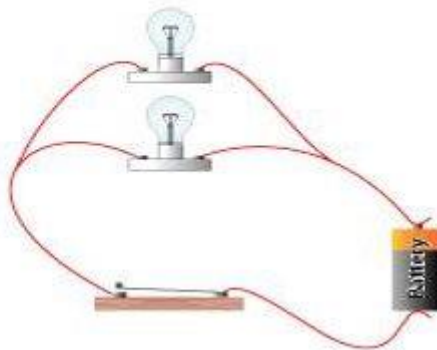


Series Circuit

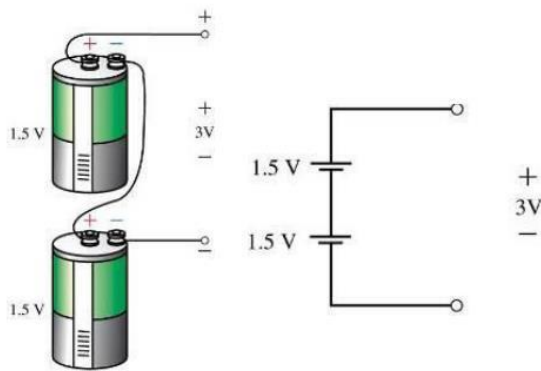
Series Circuit



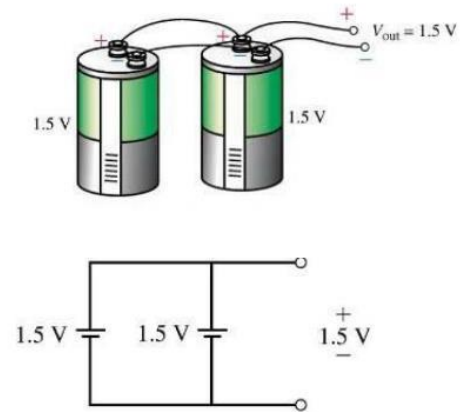
Parallel Circuit



Battery cells in series vs parallel



Cells connected in **series** increases available voltage.



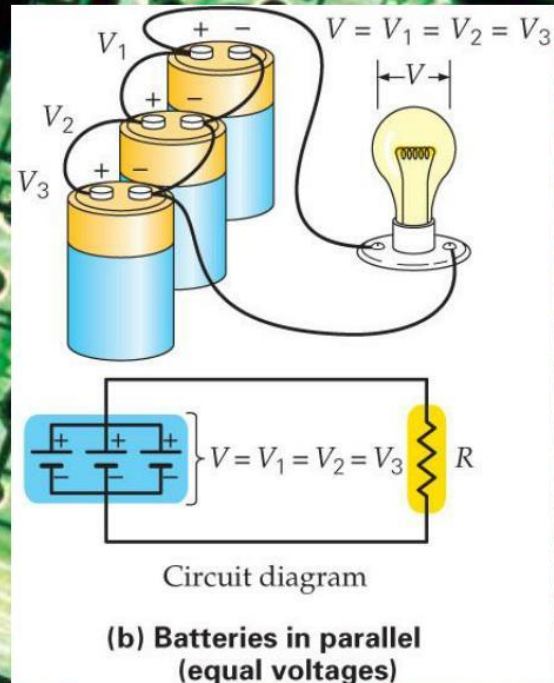
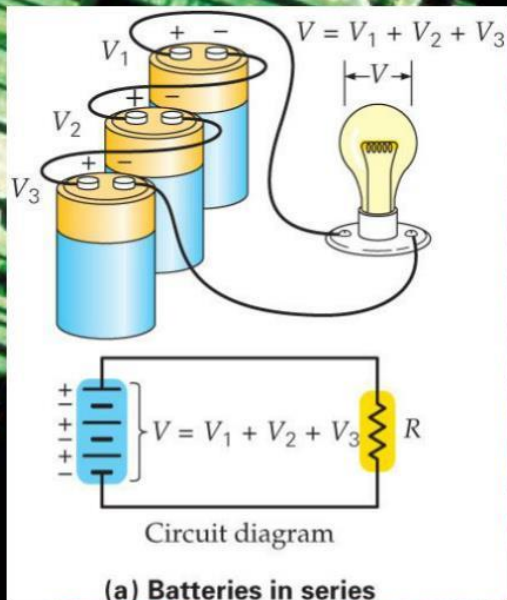
Cells connected in **parallel** increases available current.

ACTIVITY: LEARNER MUST BE ABLE TO DRAW OWN A SERIES B PARALLEL TO SHOW UNDERSTANDING (10)

A

B

Drawing Electrical Circuit Diagrams



ACTIVITY: SCENARIO (15)

You are an electrical engineer, and you need to produce a drawing for an electrical system to solve the problem and you need to use a circuit to produce either movement, light, sound, or heat, lightbulb, batteries, etc.

What is it that you need here (electrical components)

_____?

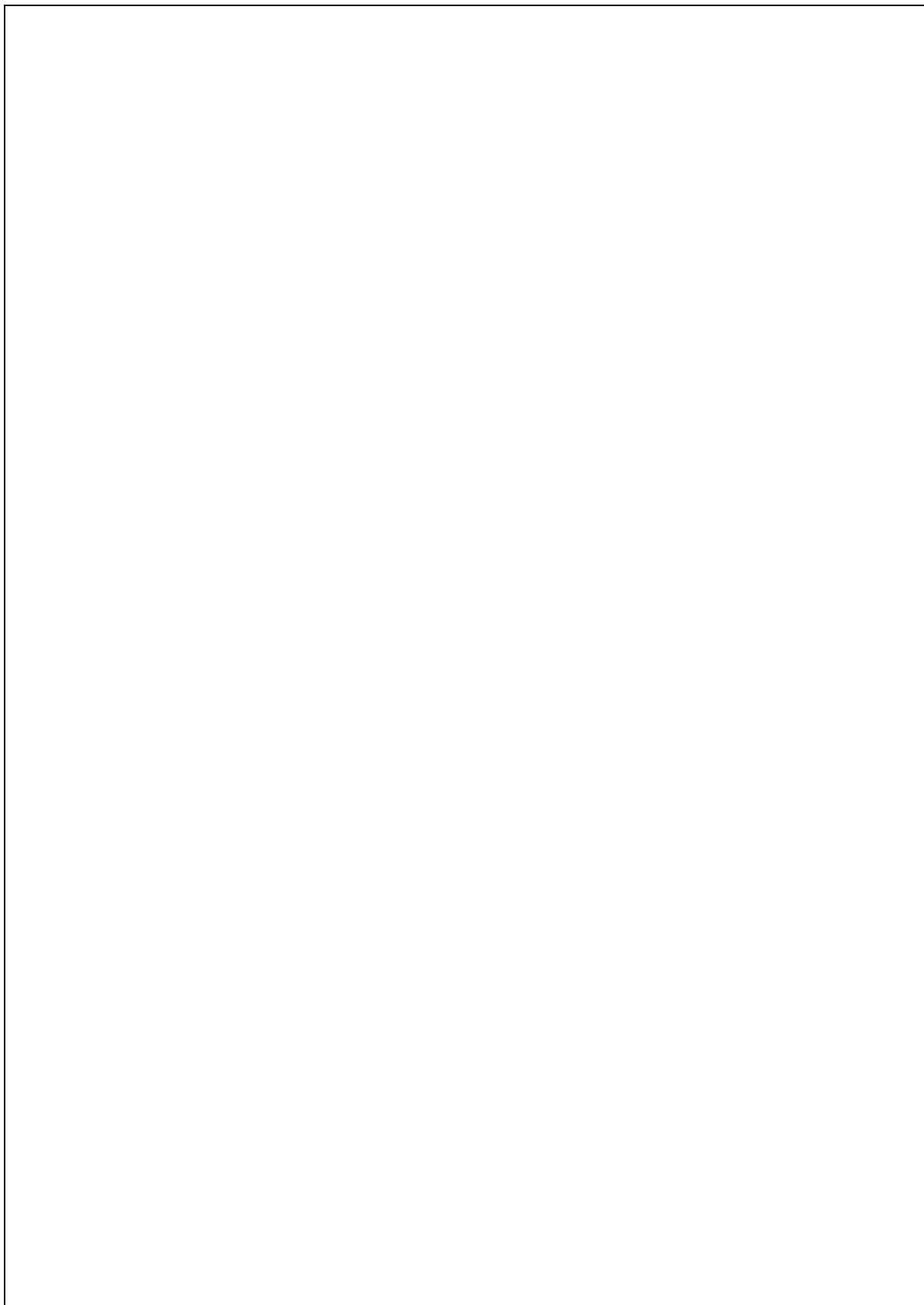
_____ , _____

_____ , _____

The following must be included

- Use electric circuits
- Produce movement, sound, light/heat
- Circuits must make use of components (cells, lightbulb, switches, wires etc.
- Where will you place your switch
- Present it to class
- Neatness

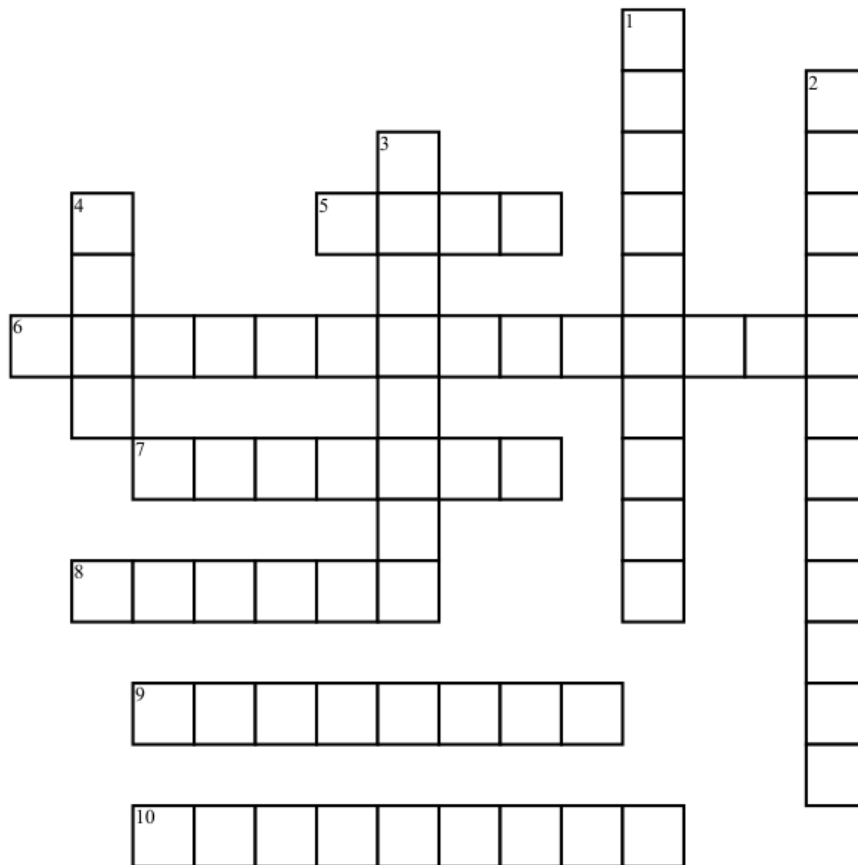
DRAWING TO FULL PG



WORD PUZZLE ON ELECTRICITY COMPONENTS

Name: _____ Date: _____

COMPONENTS OF A CIRCUIT



Across

- 5. to provide electrical energy
- 6. Connects the components together.
- 7. To measure current strength.
- 8. converts electrical energy into sound energy
- 9. a resistors whose resistance can be adjusted higher or lower.

10. to measure voltage or potential difference

Down

- 1. stops the flow of current
- 2. allows the flow of current
- 3. A component that opposes or inhibits electrical current in a circuit.
- 4. to convert electrical energy into light energy

EXPANDED OPPORTUNITIES (15)

ELECTRICITY

Word Search

H	V	R	C	V	B	K	W	B	N	F	X	V	Y	O	M	T
V	G	G	G	H	G	E	E	N	A	A	C	S	O	U	M	N
U	E	F	D	P	A	R	A	L	L	E	L	C	S	B	N	S
M	V	O	I	M	P	R	S	G	R	S	F	E	A	V	L	P
L	I	C	I	T	E	N	G	A	M	O	R	T	C	E	L	E
C	T	L	Z	K	F	T	O	E	P	I	T	T	V	I	H	Y
I	I	N	C	E	I	N	B	J	E	E	J	S	I	I	X	G
R	S	O	P	O	L	E	S	S	R	E	T	G	I	W	T	O
C	O	R	P	K	A	R	K	Y	Q	E	N	Z	M	S	J	P
U	P	T	F	Z	M	R	W	W	R	W	E	S	F	O	E	C
I	D	C	O	O	E	U	J	M	G	J	G	K	N	F	F	R
T	P	E	Y	A	N	C	I	R	O	T	A	L	U	S	N	I
S	K	L	U	V	T	N	Z	C	G	X	T	W	B	Z	L	U
M	M	E	J	X	A	L	T	L	J	J	I	M	S	G	O	C
H	L	J	N	L	J	S	H	N	I	X	V	F	G	Y	L	J
P	M	H	O	P	X	E	E	O	Q	Z	E	S	F	H	D	C
W	A	D	R	Y	C	E	L	L	S	S	C	F	L	Y	U	G

ELECTROMAGNETIC

BATTERY	ELECTRON	POLES
CHARGE	FILAMENT	POSITIVE
CIRCUIT	INSULATOR	RESISTOR
CURRENT	NEGATIVE	SERIES
DRY CELL	PARALLEL	TERMINAL

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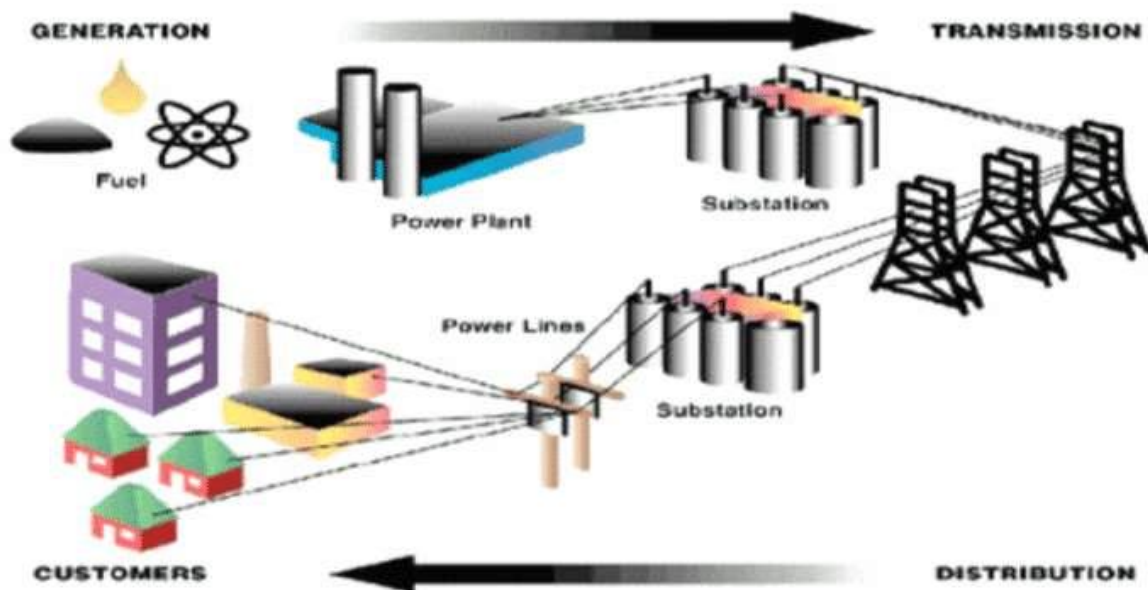
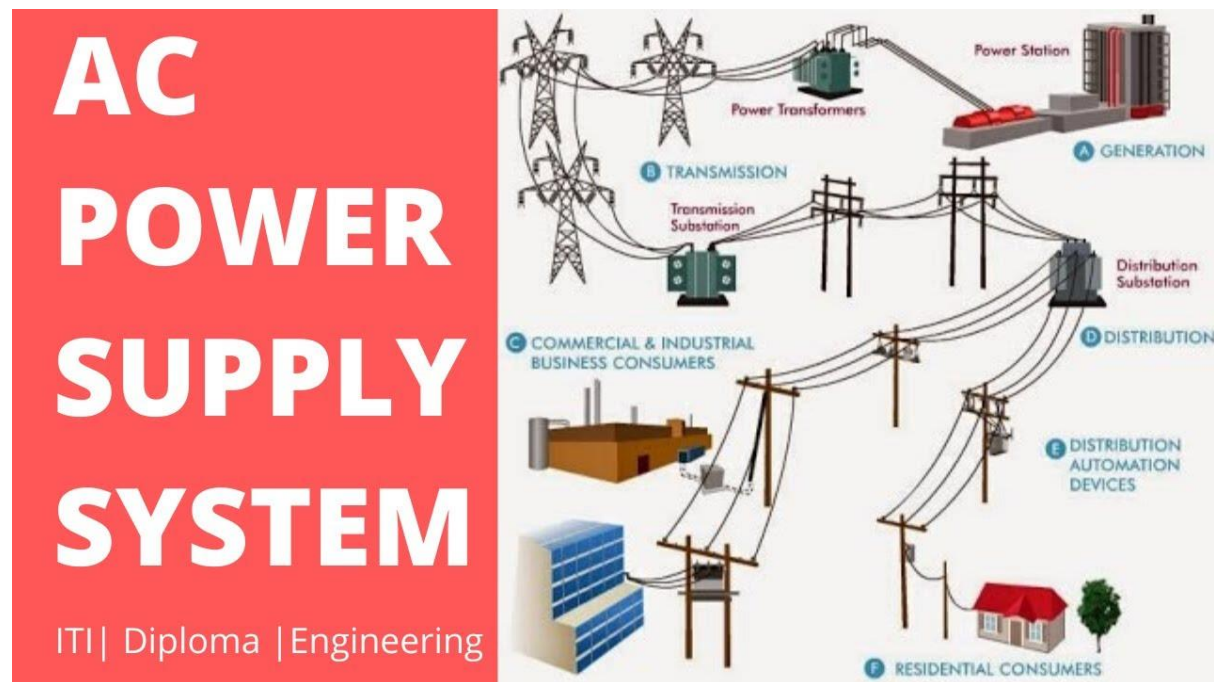
C.MAINS ELECTRICITY

THE FOLLOWING CONCEPTS ARE EXPLAINED

1.ELECTRIC SUPPLY

2.GENERATOR

3.TURBINE

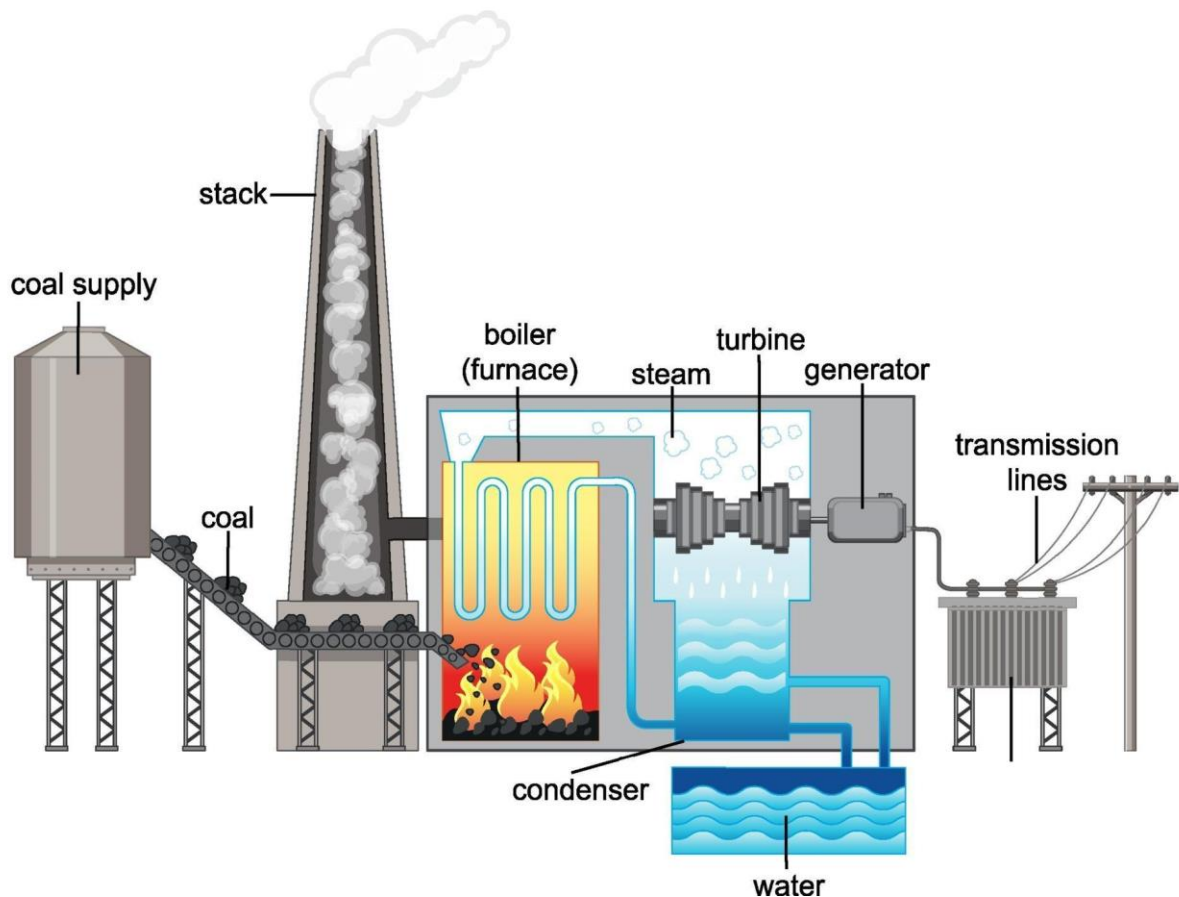


ELECTRIC SUPPLY: Means conductors used to transmit electric energy and their necessary supporting structures

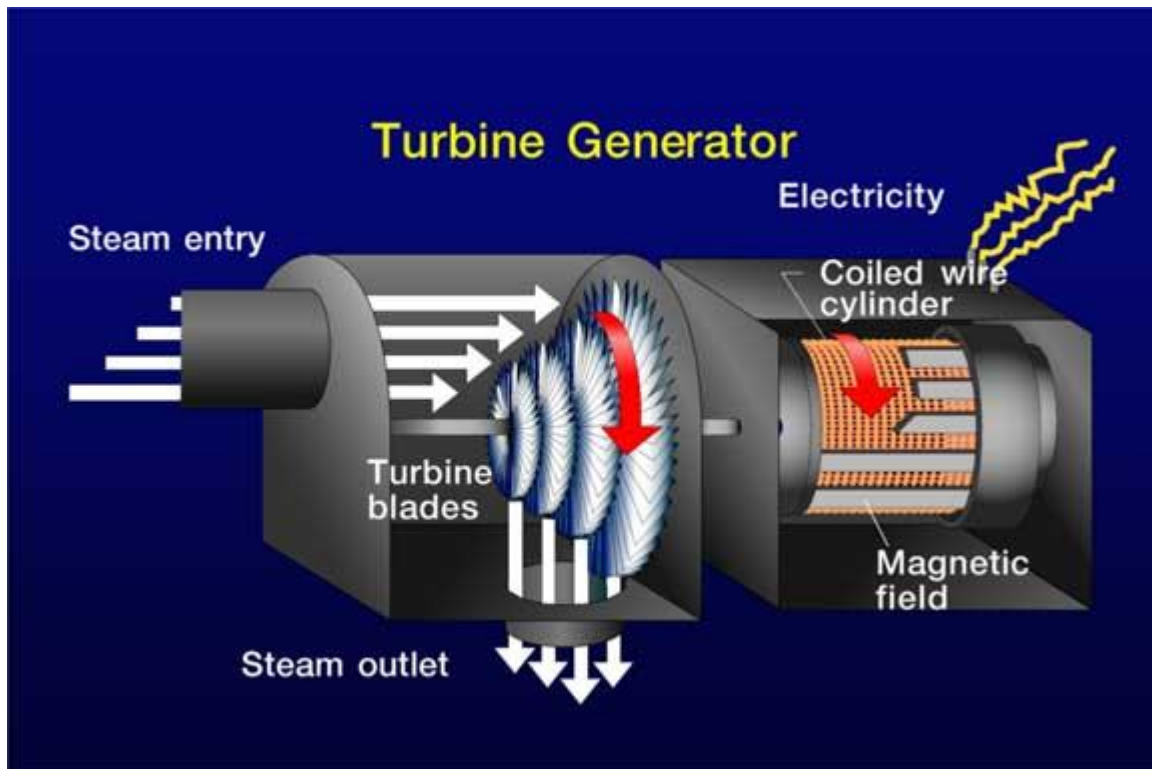
POWER SUPPLY: An electrical device that supplies electric power to electrical load: main purpose of power supply is to convert electric current from source to correct voltage

FOSSIL FUEL: Are obtained from remains of dead plants and animals from millions of years ago. coal, oil, and natural gas are example of fossil fuel

Coal Power Electrical Station

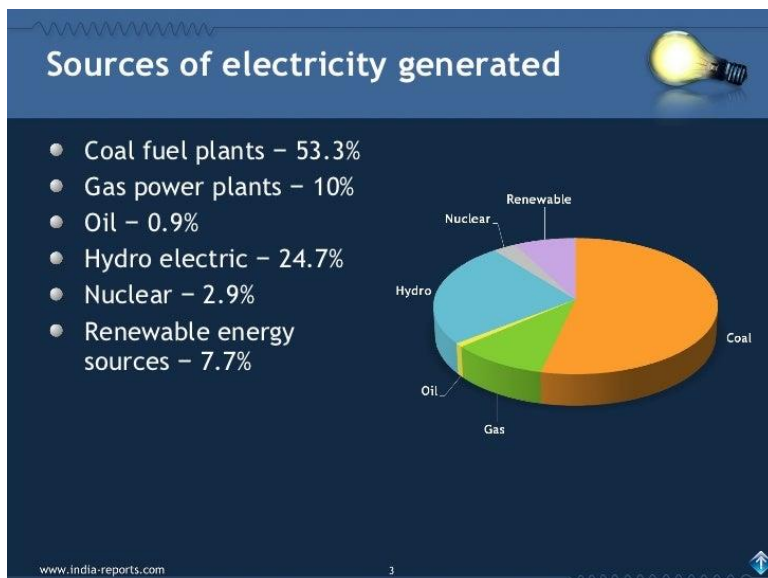


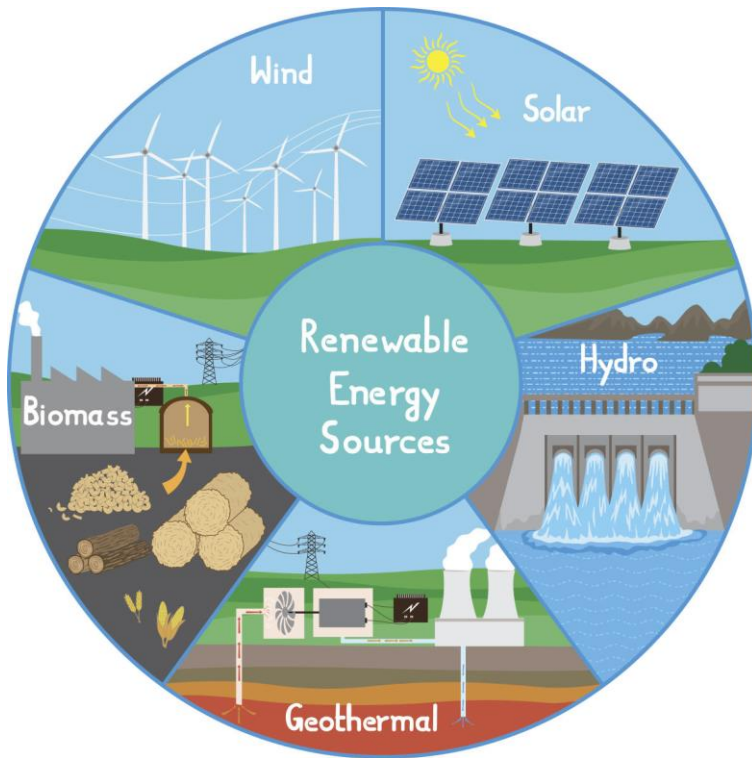
RENEWABLE ENERGY RESOURCE: They often referred to as clean energy comes from natural sources



NON RENEWABLE ENERGY: A source of energy that will eventually run out most are coal,gas and oil

A.PICTURE





ACTIVITY: LEARNERS CAN GIVE EXAMPLES OF ENERGY RESOURCES THAT ARE USED TO PRODUCE ELECTRICITY

1.	4.
2.	5.
3.	6.

(6)

**ACTIVITY: FROM (A) PICTURE DESIGN A GRAPH
ALLOCATE% USE THIS SPACE (10)**

Ancient forests turn into fossil fuels over time as the plant material accumulates and is covered with sand, clay, minerals, and rock¹. The carbon-rich organic matter is then pressed down by the covering materials, ultimately changing into one of the fossil fuels¹. As the plant

material sinks to the bottom of wet areas, it partially decays and turns into peat². With heat, pressure, and time, the peat transforms into coal². Crushed and rotted plant matter over time becomes hydrocarbons (i.e. fossil fuels)³

When those pre historic plants and animal died their bodies decomposed just in the same way as organisms decomposed today. The dead organisms became buried under layers and layers of mud, rock, sand, and water.

These layers build up and became very deep and they pushed down with a great pressure on the layers below it.

The dead plants and animals slowly decomposed and form FOSSIL FUEL

Different types of fossil fuel were formed depending on different factors depending on temperature and pressure condition during the decay of the organism. They are Oil, Natural gas and coal. The focus will be on COAL which is used to generate electricity.

COAL

In south Africa most of the power stations use coal as source of energy to generate electricity. It is mined out of the earth and transported to power station in large trucks or trains.

Coal is black rock that can be burnt to produce energy in power station all over the world. The energy in coal original comes from the energy in the Sun.

QUESTIONS

1.Different types of fossil fuel were formed on different factors name 3

2.What are these layers and layers

3.Wher is this coal from

--

4.Mode of transport they use to transport this coal

5. Where does this coal get energy from?

--

6. Explain the following in examples

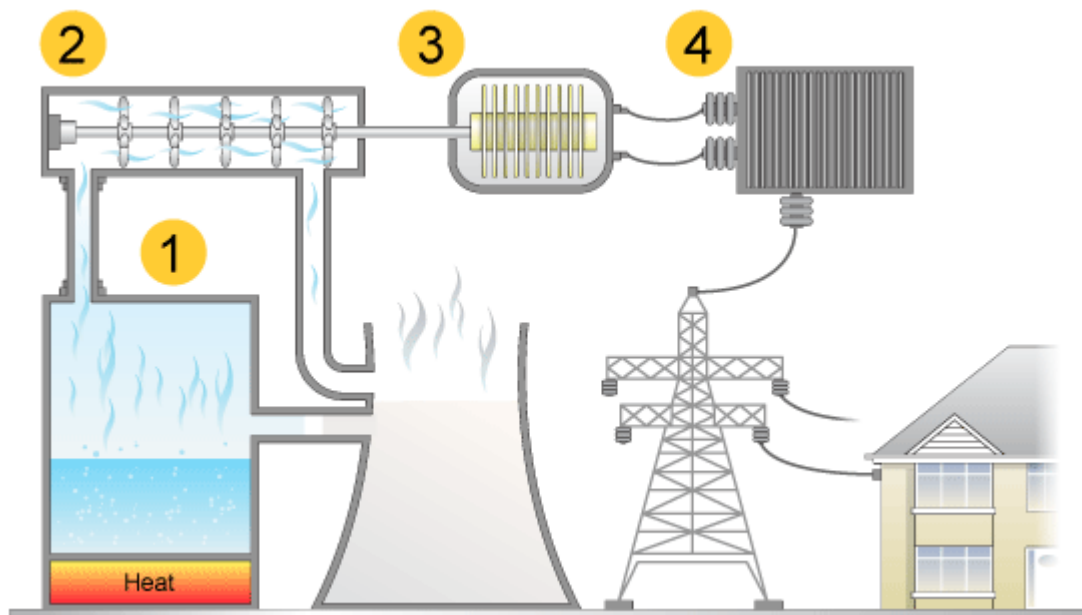
Non-Renewable

Renewable

7. Advantage and disadvantage of using wind as a source of energy to generate electricity

8. Advantage and disadvantage of using a coal to generate electricity in Eskom power stations

DIAGRAM TO BE LABELED 1-4 TO SHOW HOW ELECTRICITY IS GENERATED



(20)

ACTIVITY: INVESTIGATE HOW WE USE ALTERNATIVE IN THE OLDEN DAYS ENERGY SOURCES INSTEAD OF ELECTRICITY (10)



PASTE YOUR OWN

A large empty rectangular box for pasting a drawing or image.

GOOD LUCK
END OF TERM 3